

# CHAPTER-1

## BASICS OF INFORMATION TECHNOLOGY

### Q.1 FILL IN THE BLANKS

1. **Information Technology** has a great influence on all aspects of life.
2. **Information technology** encompasses all of the technology that we use in order to create, collect, process, protect and store information
3. **Information technology** refers to hardware, software (computer programs), and computer networks”.
4. **Information and Communication Technology** concept involves transfer of all kinds of information.
5. **I.C.T** stands for **Information and Communication Technology**.
6. **Data** can be any character, text, word, number or raw facts.
7. **Information** is data formatted in a manner that allows it to be utilized by human beings in some significant way.
8. An electronic device which accepts input from the user, processes it according to the instructions given to it & gives the required result in the form of output, is a **com-puter**.
9. A **computer** can process data, images, audio, video and graphics.
10. **Computer Architecture** is a specification detailing of how software and hardware technology standards interact to form a computer system.
11. **Computer Architecture** refers to how a computer system is designed & how it works
12. **Input Unit** helps users to enter data and commands into a computer.
13. **Arithmetic and logic unit** of the CPU performs arithmetic operations.
14. **Control Unit** is the back bone of the computers.
15. **Control Unit** is responsible for coordinating tasks between all components of a computer system.
16. **Control Unit** collects data from input units and sends it to processing units depending on its nature.
17. The **memory unit** uses a set of pre-programmed instructions to further transmit this data to other parts of CPU.
18. The third and final component of a computer system is the **output unit**.
19. **Primary Memory** is also called the internal memory of the computer.
20. **Primary Memory** holds the data and instructions on which the computer is currently working.
21. **RAM** stands for Random Access Memory.
22. **RAM** is also known as read/write memory.
23. **RAM** is also called volatile memory.
24. **ROM** stands for read only memory.
25. **ROM** is also known as Read Only Memory.
26. **Bit** is a binary digit that holds only one or two values.
27. **Nibble** is a group of 4 bits.
28. **Byte** is a group of 8 bits.
29. **Byte** is the smallest character that can represent a data item or a character.

30. **Computer Hardware** comprises of the physical components that a computer system requires to function.
31. **Computer Hardware** consists of physical components that a computer system requires to function.
32. A set of instructions given to a computer is known as a **program**.
33. **Software** is a generic term used to describe computer programs.
34. **Open source software** refers to the software which releases code in public domain for anyone to use it.
35. **FOSS** means Free Open Source Software Policy.
36. **Application software** is a program or a group of programs designed for end users.
37. **System software** is a set of instructions required for a computer to work.
38. **Linux Operating system** is a program that allows different applications and various components of hardware such as monitor, printer, mouse, keyboard etc. to communicate with each other.
39. **DOS (Disk Operating System)** was dominant operating system for microcomputers in the 1980's and early 1990's.
40. **Chrome OS** is an open-source operating system created by Google.
41. Apple Corporation's registered operating system is called **Mac OS**.
42. **UNIX** was trademarked in 1969 by a group of **AT&T employees** at Bell Labs.
43. **Linux** is an open-source, portable, multi-user, multiprogramming operating system.
44. **Blackberry** is a closed source operating system for smartphones and tablet devices.
45. **Android** is a free and open source operating system provided by Google.
46. **Asha** is used by Nokia phones. This is closed source OS.
47. **iOS** is Apple's closed source operating system for Apple's iPhone, iPod Touch, iPad and second-generation Apple TVs.
48. **Windows Phone** is developed by Microsoft as a closed source operating system for mobile phones.
49. **GNU/Linux** is a family of operating systems that are created by various companies, organizations and individuals.
50. **GNU/LINUX** is created by using Free Software Philosophies.
51. **Ubuntu** is the most popular GNU/LINUX distribution.
52. GUI is also called as **GNOME**.
53. The **command line interface** is when the user interacts with the computer using text.
54. Commands are typed into a **terminal**.
55. The default CLI on Ubuntu is called **bash**.
56. All files and directories are stored under the **root directories**.
57. Essential system programs are stored **/bin**.
58. All connected devices are stored under **/dev**.
59. System configuration files are stored under **/etc**.
60. **/proc** provides information about processes.
61. Temporary files are stored under **/tmp**.
62. User directories are stores under **/home**.
63. A **computer network** consists of collection of computers, printers and other equipment that is connected together so that they can communicate with each other.
64. **Network architecture** is the design of a computer network.

65. **Network architecture** is a framework for the specification of a network's physical components and their functional organization and configuration.
66. A **Peer-to-Peer network** has no dedicated servers.
67. In peer to peer architecture all the machines are called **peers**.
68. **Client** is a computer which is seeking any resource from another computer.
69. **Server** is a computer that has a resource which is served to another computer.
70. **Internet** means connecting computer to another computer anywhere in the world.
71. **Internet** is a highway of information.
72. The **internet** is generally defined as a global network connecting millions of computers.
73. **ARPANET** means Advanced Research Projects Agency Network.
74. **ARPANET** uses packet switching to allow multiple computers to communicate on a single computer.
75. **Tim Berners-Lee** invented the World Wide Web (WWW).
76. A **protocol** is a set of rules that governs the communications between the computers on a network.
77. **TCP/IP** stands for Transmission Control Protocol & Internet Protocol.
78. **Domain Name System** translates network address (such as IP addresses) into terms understood by humans (such as Domain Names) and vice-versa.
79. **Dynamic Host Configuration Protocol** can automatically assign internet addresses to computers and users.
80. **File Transfer Protocol** is a protocol that is used to transfer and manipulate files on the internet.
81. **Hyper Text Transfer Protocol** is an internet-based protocol for sending and receiving web pages.
82. **Internet Message Access Protocol** is a protocol for receiving e-mail messages from server on the Internet.
83. **Internet Relay Chat** is a protocol used for Internet chat and other communications.
84. **Post Office Protocol Version 3** is a protocol used for receiving e-mail from remote mail servers.  
**Post Office Protocol Version 3** does not maintain copy on the server
85. **Simple Mail Transfer Protocol** is a protocol for sending e-mail messages to the Server on the Internet.
86. **Green Computing** is the study and practice of environmentally sustainable computing or IT.
87. The **Internet of Things (Iot)** is the network of physical devices, vehicles, home appliances, and other items embedded with electronics, software, sensors, actuators, and connectivity which enables these things to connect, collect and exchange data.
88. **Cloud Computing** is the delivery of computing services - servers, storage, databases, networking, software, data analytics and more - over the internet.
89. **Data Analytics (DA)** is the process of examining data sets in order to draw conclusions about the information they contain, increasingly with the help of specialized systems and software.
90. **Artificial Intelligence (AI)** is intelligence demonstrated by machines, in contrast to

the natural intelligence displayed by humans and other animals.

91. **Machine Learning (ML)** is the scientific study of algorithms and statistical models that computer systems use to effectively perform specific task without using explicit instructions, relying on patterns and inference instead.  
**Machine Learning (ML)** is seen as a subset of artificial intelligence.
92. **Big Data** refers to data sets that are too large or complex for traditional data-processing application software to adequately deal with.
93. **Block chain** is a growing list of records, called blocks, which are linked using cryptography.
94. **LAN** covers smaller geographical area (Size is limited to a few kilometres) and are privately owned.
95. **MAN** is larger area than that of a LAN and smaller area as compared to WAN.
96. **Wide Area Network** is a computer network that extends over a large geographical area.
97. An **operating system** is a software program that empowers the computer hardware to communicate and operate with the computer software.
98. The **operating system** boots up the computer and makes sure that everything is operational.
99. The **operating system** is also what runs the cell phone and most of the electronic devices.
100. The **operating systems** usually come preloaded on any computer, tablet, laptop or cell phone that you buy.
101. **IT Enabled Services** (ITES), also called web enabled services or remote services.
102. The **graphical user interface** is when the user interacts with the computer using images, icons, and dialog boxes.
103. The **command line interface** is when user interacts with the computer using text.
104. **Closed Source software** means that the software is used by the proprietary and has a closely guarded code.
105. **Computer software** can be classified into two types based on its utility
106. **Computer storage** & memory is often measured in Megabytes and Gigabytes.
107. **Open Source Software** is a code free software available freely to copy and modify.
108. **Closed source Software** is having proprietary rights with the developers & are paid ones.
109. **GNU/Linux** operating systems are usually free of charge, free to distribute, and they are open source.
110. **Internet** means connecting computer to any other computer anywhere in the world.
111. **List Directories** Prints the list of directories in the current one.
112. **Print Working Directory** Prints the path to the current directory.
113. **Who Am I?** Prints the name of the current user on terminal.
114. **List CPU** Prints information about your CPU on terminal.
115. **Touch** Creates a new file.
116. **Change Directory** Control goes to a different directory.
117. **Nano** is a text editor.
118. **Concatenate** Prints the contents of the files listed on the screen.
119. **Echo** is used to display line of text or string that are passed as an argument.
120. **Clear** is used to display line of text or string that are passed as an argument

## Q.2 STATE TRUE OR FALSE

1. Nibble is a binary digit that holds two or more values. -- **FALSE**
2. Bit is a binary digit that holds only one of two values. -- **TRUE**
3. A group of 4 bits is called a byte. -- **FALSE**
4. A group of 8 bits is called a byte. -- **TRUE**
5. A group of 4 bits is called nibble. -- **TRUE**
6. A computer would not be able to function properly without an operating system. -- **TRUE**
7. Application software is a software program that empowers the computer hardware to communicate and operate with the computer software. -- **FALSE**
8. An operating system boots up the computer and makes sure that everything is operational. -- **TRUE**
9. The application software is also what runs the cell phones and most of the electronic devices. -- **FALSE**
10. Intranet is a highway of information. -- **FALSE**
11. Information Communication Technology has a great influence on all aspects of life. **FALSE**
12. IT (Information technology) encompasses all of the technology that we use in order to create, collect, process, protect and store information. **TRUE**
13. ICT (Information and Communication Technology) concept involves transfer of all kinds of information. **TRUE**
14. Data can be any character, text, word, number or raw facts. **TRUE**
15. Information is data formatted in a manner that allows it to be utilized by human beings in some significant way. **TRUE**
16. A computer can process data, images, audio, video and graphics. **TRUE**
17. Computer Architecture is a specification detailing of how software and hardware technology standards interact to form a computer system. **TRUE**
18. Computer Architecture refers to how a computer system is designed and how it works. **TRUE**
19. Control Unit helps users to enter data and commands into a computer. **FALSE**
20. Arithmetic and logic unit of the CPU performs arithmetic operations. **TRUE**
21. Memory Unit is the back bone of the computers. **FALSE**
22. Memory Unit is responsible for coordinating tasks between all components of a computer system, **FALSE**
23. Memory Unit collects data from input units and sends it to processing units depending on its nature. **FALSE**
24. The Memory unit uses a set of per-programmed instructions to further transmit this data to other parts of CPU. **TRUE**
25. Secondary Memory is also called the internal memory of the computer. **FALSE**
26. Primary Memory holds the data and instructions on which the computer is currently working. **TRUE**
27. RAM stands for Random Access Memory. **TRUE**
28. ROM is also known as read/write memory. **FALSE**
29. RAM is also called volatile memory. **TRUE**

30. ROM stands for read only memory. **TRUE**
31. ROM is also known as Read Only Memory. **TRUE**
32. Bit is a binary digit that holds only one or two values. **TRUE**
33. Nibble is a group of 4 bits. **TRUE**
34. Byte is a group of 8 bits. **TRUE**
35. BIT is the smallest character that can represent a data item or a character. **FALSE**
36. Computer Hardware comprises of the physical components that a computer system requires to function. **TRUE**
37. Computer Software consists of physical components that a computer system requires to function. **FALSE**
38. A set of instructions given to a computer is known as a program. **TRUE**
39. Software is a generic term used to describe computer programs. **TRUE**
40. Open source software refers to the software which releases code in public domain for anyone to use it. **TRUE**
41. FOSS means Free Open Source Software Policy. **TRUE**
42. System software is a program or a group of programs designed for end users. **FALSE**
43. System software is a set of instructions required for a computer to work. **TRUE**
44. DOS (Disk Operating System) was dominant operating system for microcomputers in the 1980's and early 1990's. **TRUE**
45. Chrome OS is an open-source operating system created by Google. **TRUE**
46. LINUX was trademarked in 1969 by a group of AT&T employees at Bell Labs. **FALSE**
47. Linux is an open-source, portable, multi-user, multiprogramming operating system. **TRUE**
48. Blackberry is a closed source operating system for smartphones and tablet devices. **TRUE**
49. GNU/Linux is a family of operating systems that are created by various companies, organizations and individuals. **TRUE**
50. GNU/LINUX is created by using Free Software Philosophies. **TRUE**
51. Ubuntu is the most popular GNU/LINUX distribution. **TRUE**
52. CUI is also called as GNOME. **FALSE**
53. The command line interface is when the user interacts with the computer using text. **TRUE**
54. Commands are typed into a terminal. **TRUE**
55. The default CLI on Ubuntu is called bash, **TRUE**
56. All files and directories are stored under the root directories. **TRUE**
57. Essential system programs are stored /bin. **TRUE**
58. All connected devices are stored under /dev. **TRUE**
59. System configuration files are stored under /etc. **TRUE**
60. /proc provides information about processes . **TRUE**
61. Temporary files are stored under /tmp. **TRUE**
62. User directories are stored under /home. **TRUE**
63. A computer network consists of collection of computers, printers and other equipment that is connected together so that they can communicate with each other. **TRUE**
64. A Peer-to-Peer network has dedicated servers. **FALSE**
65. In peer to peer architecture all the machines are called peers. **TRUE**
66. Client is a computer which is seeking any resource from another computer. **TRUE**



67. Server is a computer that has a resource which is served to another computer. **TRUE**
68. Internet means connecting computer to another computer anywhere in the world. **TRUE**
69. Internet is a highway of information. **TRUE**
70. The internet is generally defined as a global network connecting millions of computers. **TRUE**
71. ARPANET means Advanced Research Projects Agency Network. **TRUE**
72. ARPANET uses packet switching to allow multiple computers to communicate on a single computer. **TRUE**
73. Tim Berners-Lee invented the World Wide Web (WWW). **TRUE**
74. A protocol is a set of rules that governs the communications between the communications between computers on a network. **TRUE**
75. TCP/IP stands for Transmission Control Protocol & Internet Protocols. **FALSE**
76. Domain Naming System translates network address (such as IP addresses) into terms understood by humans (such as Domain Names) and vice-versa. **FALSE**
77. Dynamic Host Configuration Protocol can automatically assign internet addresses to computers and users. **TRUE**
78. File Transfer Protocol is a protocol that is used to transfer and manipulate files on the internet. **TRUE**
79. Hyper Text Transfer Protocol is an internet-based protocol for sending and receiving web pages. **TRUE**
80. Internet Message Access Protocol is a protocol for receiving e-mail messages from server on the Internet. **TRUE**
81. Internet Relay Chat is a protocol used for Internet chat & other communication. **TRUE**
82. Post Office Protocol Version 3 is a protocol used for receiving e-mail from remote mail servers. **TRUE**
83. Post Office Protocol Version 3 does not maintain copy on the server. **TRUE**
84. Simple Mail Transfer Protocol is a protocol for sending e-mail messages to the Server on the Internet. **TRUE**
85. Green Computing is the study and practice of environmentally sustainable computing or IT. **TRUE**
86. The Internet of Things (IoT) is the network of physical devices, vehicles, home appliances, and other items embedded with electronics, software, sensors, actuators, and connectivity which enables these things to connect, collect and exchange data. **TRUE**
87. Cloud Computing is the delivery of computing services - servers, storage, databases, networking, software, data analytics and more - over the internet. **TRUE**
88. Data Analytics (DA) is the process of examining data sets in order to draw conclusions about the information they contain, increasingly with the help of specialized systems and software. **TRUE**
89. Artificial Intelligence (AI) is intelligence demonstrated by machines, in contrast to the natural intelligence displayed by humans and other animals. **TRUE**
90. Machine Learning (ML) It is the scientific study of algorithms and statistical models that computer systems use to effectively perform specific task without using explicit instructions, relying on patterns and inference instead. **TRUE**

91. Big Data refers to data sets that are too large or complex for traditional data-processing application software to adequately deal with. **TRUE**
92. Block chain is a growing list of records, called blocks, which are linked using cryptography. **TRUE**
93. WAN covers smaller geographical area (Size is limited to a few kilometres) and are privately owned. **FALSE**
94. MAN is larger area than that of a LAN and smaller area as compared to WAN. **TRUE**
95. Wide Area Network is a computer network that extends over a large geographical area. **TRUE**
96. An operating system is a software program that empowers the computer hardware to communicate and operate with the computer software. **TRUE**
97. The operating system boots up the computer and makes sure that everything is operational. **TRUE**
98. The operating system is also what runs the cell phone and most of the electronic devices. **TRUE**
99. The operating systems usually come preloaded on any computer, tablet, laptop or cell phone that you buy. **TRUE**
100. IT Enabled Services (ITES), also called web enabled services or remote services. **TRUE**
101. The graphical user interface is when the user interacts with the computer using images, icons, and dialog boxes. **TRUE**
102. The commands are programs that are run when user types command name. **TRUE**
103. The command line interface is when user interacts with the computer using text. **TRUE**
104. Closed Source software means that the software is used by the proprietary and has a closely guarded code. **TRUE**
105. Computer software can be classified into two types based on its utility. **TRUE**
106. Computer storage & memory is often measured in Megabytes and Gigabytes. **TRUE**
107. Open Source Software is a code free software available freely to copy & modify. **TRUE**
108. Closed source Software is having proprietary rights with the developers & are paid ones. **TRUE**
109. GNU/Linux operating systems are usually free of charge, free to distribute, and they are open source. **TRUE**
110. Internet means connecting computer to any other computer anywhere in the world. **TRUE**
111. List Directories Prints the list of directories in the current one. **TRUE**
112. Print Working Directory Prints the path to the current directory. **TRUE**
113. Who Am I? Prints the name of the current user on terminal. **TRUE**
114. List CPU Prints information about your CPU on terminal. **TRUE**
115. Touch Creates a new file. **TRUE**
116. Change Directory Control goes to a different directory. **TRUE**
117. Nano is a text editor. **TRUE**
118. Concatenate Prints the contents of the files listed on the screen. **TRUE**
119. Echo is used to display line of text or string that are passed as an argument. **TRUE**
120. Clear is used to display line of text or string that are passed as an argument. **TRUE**



### Q.3 CHOOSE THE CORRECT SINGLE ALTERNATIVE

1. In GNU/LINUX the topmost directory is called the root directory.  
a) root directory b)/tamp directory c)/etc directory d)/proc directory
2. The root directory is written as /  
a)/home b)/etc c)/bin d)/
3. Extranet is a combination of internet and ARPANET  
a) extranet b) server c) internet d) software
4. IT enables services are also called web enabled services  
a) operating systems b) GUI c) Internet d) web enabled services
5. Information security analysts are responsible for ensuring that networks are watertight  
a) web designer b) database manager c) Financial advisor d) Cyber advisor
6. The full form of TCP/IP is Transmission Control Protocol and Internet Protocol  
a) Transmission Control Protocol and Internet Protocol  
b) Transfer Control Protocol and Internet Protocol  
c) Transmission Control Protocol and Internet Protocols  
d) Transmission Control Protocols and Internet Protocol
7. DNS stands for Domain Name System.  
a) Dedicated Name system b) Domain Name System c) Domain Name Systems  
d) Dedicated Name systems
8. DNS translates network address into terms understood by humans  
a) DNS b) DHCP c) FTP d) IMAP
9. The full form of DHCP is Dynamic Host Configuration Protocol.  
a) Dynamic Host Configuration Protocol  
b) Dedicated Host Configuration Protocol  
c) Dynamic Host Configuration Protocols  
d) Dedicated Host Configuration Protocols
10. FTP means File Transfer Protocol.  
a) File Transfer Protocol  
b) File Transport Protocol  
c) File Transferring Protocol  
d) File Transfer Protocols

11.FTP is used to transfer and manipulate files on the internet

- a) IRC b) POP3 c) SMTP d) FTP

12.HTTP stands for Hypertext Transfer Protocol.

- a) Hypertext Transferred Protocol
- b) Hypertext Transfer Protocols
- c) Hypertext Transferring Protocol
- d) Hypertext Transferring Protocols

13.DHCP stands for Dynamic Host Configuration Protocol.

- a) Dynamic Host Configuration Protocols
- b) Dynamic Host Configuration Protocol
- c) Dedicated Host Configuration Protocol
- d) Dedicated Host Configuration Protocols

14.IMAP stands for Internet Message Access Protocol.

- a) Internet Message Access Protocols
- b) Internet Message Access Protocol
- c) Internet Messaging Access Protocol
- d) Internet Message Accessing Protocol

15.IRC stands for Internet Relay Chat.

- a) Internet Relay Chat
- b) Intranet Relay Chat
- c) International Relay Chat
- d) Internal Relay Chat

16.POP3 stands for Post Office Protocol Version 3.

- a) Pre Office Protocol Version 3
- b) Post Office Protocol Version 3
- c) Primary Office Protocol Version 3
- d) Physical Office Protocol Version 3

17.SMTP stands for Simple Mail Transfer Protocol.

- a) Sample Mail Transfer Protocol
- b) Synchronous Mail Transfer Protocol
- c) Simpler Mail Transfer Protocol
- d) Simple Mail Transfer Protocol

18.SMTP means a protocol for sending message to the server on the internet

- a) POP3 b) SMTP C) IMAP d) HTTP

19.POP3 is a protocol used for receiving email from remote mail servers. It does not maintain copy on the server

- a) IRC b) IP C) IMAP d) POP3

20. IRC is a protocol used for internet chat and other communications  
a) FTP b) IMAP c) IRC d) DNS
21. IRC facilitates communication in the form of text  
a) FTP b) IRC c) DHCP d) DNS
22. IMAP is a protocol for receiving email messages from server on the internet  
a) IMAP b) POP3 c) SMTP d) DNS
23. IMAP maintains copy of all emails on the server  
a) IRC b) POP3 c) SMTP d) IMAP
24. HTTP is an internet based protocol for sending and receiving web pages  
a) IRC b) HTTP c) SMTP d) DNS
25. FTP is a protocol that is used to transfer and manipulate files on the internet  
a) TCP b) HTTP c) FTP d) SMTP
26. DHCP can automatically assign internet addresses to computers and user  
a) DHCP b) HTTP c) IMAP d) SMTP
27. DNS translates network address (such as IP address) into terms understood by humans (such as Domain names) and vice versa  
a) POP3 b) SMTP c) DNS
28. TCP/IP breaks down the message into packets and send them out into the network  
a) TCP b) HTTP c) FTP d) SMTP
29. A protocol is a set of rules that governs the communication between computers on a network  
a) software b) protocol c) Internet d) Network
30. Green computing is the study and practice of environmentally sustainable computing or IT  
a) cloud computing b) Data analysis c) Big data d) green computing
31. Big data refers to data sets that are too large or complex for traditional data processing application software to adequately deal with.  
a) cloud computing b) Data analysis c) Big data d) green computing
32. Block chain is a growing list of records called blocks which are linked using cryptography  
a) Artificial intelligence b) block chain c) Bid data d) Machine learning

33. Cloud computing is the delivery of computing services – servers, storage databases, networking software, data analytics and more over the internet  
a) cloud computing b) Data analysis c) Big data d) green computing
34. Bit is a binary digit that holds only one of the two value 0 or 1  
a) Bit b) Nibble c) Byte d) Kilobyte
35. MS DOS is an example of CLI  
a) MS DOS b) chrome so c) windows d) windows
36. The default CLI on Ubuntu is called bash  
a) Android b) IOS c) bash d) blackberry
37. Essential system programs are stored here /bin  
a) /bin b) /proc c) /etc d) /home
38. All connected devices are stored here. Including internal devices, temperature sensors, and batteries /dev  
a) /dev b) /proc c) /etc d) /home
39. System configuration files are stored here /etc.  
a) /bin b) /proc c) /etc d) /home
40. Files that provide information about processes are stored here /proc. This is information like, how much RAM is free, or how fast the CPUs are running  
a) /dev b) /proc c) /etc d) /home
41. Temporary files are stored here /tmp  
a) /tmp b) /proc c) /etc d) /home
42. LAN covers smaller geographical area and are privately owned.  
a) LAN b) WAN c) MAN d) SAN
43. Users home directories are stored here /home.  
a) /tmp b) /proc c) /etc d) /home
44. MAN is larger area than that of a LAN and smaller area as compared to WAN  
a) LAN b) WAN c) MAN d) SAN
45. Wide Area Network is a computer network that extends over a large geographical area.  
a) LAN b) WAN c) MAN d) SAN

46. The Internet of Things (Iota) is the network of physical devices, vehicles

- a) Internet of Things b) Block chain c) Big Data d) Data analytics

47. Open Source Software is a code free software available freely to copy and modify

- a) open source software b) closed source software c) hardware d) protocol

48. Closed source Software is having proprietary rights with the developers and are paid ones.

- a) open source software b) closed source software c) hardware d) protocol

#### **Q.4 CHOOSE THE CORRECT DOUBLE ALTERNATIVE:**

1. The main function of input device is to direct commands and data into computer.

2. The 2 types of computer memory are Primary Memory and Secondary Memory.

3. The 2 types of RAM are Dynamic RAM and Static RAM.

4. Computer storage is often measured in Megabytes (MB) and Gigabytes (GB).

5. UNIX is a multitasking and multi-user computer operating system.

6. The outcome of an IT enabled service is in two forms: Direct Improved Service and Indirect Benefits.

7. \_\_\_\_\_ and \_\_\_\_\_ are the examples on input devices

- a) MICR b) OCR c) printer d) monitor

Ans- **MICR** and **OCR**

8. \_\_\_\_\_ and \_\_\_\_\_ are the examples of secondary memory

- a) magnetic hard disk b) optical disk c) RAM d) ROM

Ans- **magnetic hard disk** and **optical disk**

9. \_\_\_\_\_ and \_\_\_\_\_ are the examples of primary memory

- a) RAM b) ROM c) Magnetic hard disk

Ans- **RAM** and **ROM**

10. \_\_\_\_\_ and \_\_\_\_\_ are the examples of application software

- a) Microsoft office b) Tally c) IOS d) Windows

Ans- **Microsoft office** and **tally**

11. \_\_\_\_\_ and \_\_\_\_\_ are the examples of application software

- a) Microsoft b) Tally c) IOS d) windows

Ans- **Microsoft** and **tally**

12. \_\_\_\_\_ and \_\_\_\_\_ are the examples of operating system

- a) Android b) IOS c) UNIX d) LINUX

Ans- **UNIX** and **LINUX**

13. \_\_\_\_\_ and \_\_\_\_\_ are the examples of operating system for mobile phones

a) Android b) IOS c) UNIX d) LINUX

and- **IOS** and **android**

14. Computer storage and memory is often measured in \_\_\_\_\_ and \_\_\_\_\_

a) megabytes b) kilobytes c) bytes d) gigabytes

Ans- **Megabytes (MB)** and **Gigabytes (GB)**.

15. Computer software can be classified into two types-\_\_\_\_\_ and \_\_\_\_\_

a) application software b) system software c) protocol d) software

Ans- **Application Software** and **System Software**.

## **Q.5 CHOOSE THE CORRECT TRIPLE ALTERNATIVE**

1. IT refers to **hardware, software** and **computer networks**.
2. Every computer system has three basic components: **Input Unit, Central Processing Unit & Output Unit**.
3. Examples of input unit are **Magnetic Ink Character Reader (MICR), Optical Character Reader (OCR) & Keyboard**.
4. The 3 elements of CPU are **Arithmetic and logical unit, Control Unit & Memory Unit**.
5. The 3 types of ROM are **PROM, EPROM & EEPROM**.
6. The examples of secondary storage devices are **Magnetic (Hard) Disk, Pen Drives & Optical Disk**.
7. The examples of Output Unit are **monitors, screens & speakers**.
8. In open source software the source code can be **copied, modified or distributed by other users & organizations**.
9. In closed source software only the original authors of the organization can **access, copy & alter** that software.
10. Linux is an **open-source, portable & multi-user operating system**.
11. The graphical user interface is when the user interacts with the computer using **images, icons & dialog boxes**.
12. The three types of networks are **LAN, MAN & WAN**.
13. IT Enabled Services (ITES) are also called as **web enabled services, remote services or Tele-working**.



## Q.6 MATCH THE COLUMNS

### COLUMN A

- 1.Primary memory
- 2.Secondary memory
- 3.Input unit
- 4.Output unit
- 5.Memory unit
- 6.Open source software
7. Closed source software
8. System software
9. FOSS
10. Asha
11. Linux
12. LAN
13. MAN
14. WAN
- 15.WWW

### COLUMN B

- A. Main memory
- B. Auxiliary memory
- C. keyboard
- D. Printer
- E. Cache
- F. Tally
- G. MS office
- H. Windows
- I. Liber office
- J. Nokia
- K. Ubuntu
- L. Schools
- M. ISP
- N. Satellite
- O. Tim breeder's lee

## Q.7 ANSWER THE FOLLOWING:

### 1. Define:

**1. Information Technology** - "IT (Information Technology) encompasses all of the technologies that we use in order to create, collect, process, protect and store information.

**2. Information Technology** refers to hardware, software (computer programs), and computer networks".

**3. I.C.T (Information and Communication Technology)** concept involves transfer and use of all kinds of information.

**4.I.C.T (Information and Communication Technology)** is the foundation of economy and a driving force of social changes in the 21<sup>st</sup> century.

**5. Data** - Data can be any character, text, word, number or raw facts.

**6. Information** - Information is data formatted in a manner that allows it to be utilized by human beings in some significant way.

**7. Computer Architecture** is a specification detailing of how a set of software and hardware technology standards interact to form a computer system

## **2. State the functions of Computer System.**

Ans A computer can process data, images, audio, video and graphics.

Functions of Computer system are as follows

- 1) It accepts data or instructions by way of input.
- 2) It stores data.
- 3) It can process data as required by the user.
- 4) It gives results in the form of output.
- 5) It controls all operations inside a computer.

## **3. Write a short note on following:**

### **1. Input Unit –**

- 1.**This unit helps users to enter data and commands into a computer system. Data can be in the form of numbers, words, actions, commands, etc.
- 2.** The main function of input devices is to direct commands and data into computer.
- 3.** Computer then uses its CPU to process data and produce output.
- 4.** Examples of input unit are Keyboard, Mouse, Magnetic Ink Character Reader (MICR), Optical Character Reader (OCR)

### **2. Central Processing Unit (CPU):**

- 1.** After receiving data and commands from users, a computer system has to process it according to the instructions provided. Here, it has to rely on a component called the Central Processing Unit.
- 2. Arithmetic and Logic Unit:** This part of the CPU performs arithmetic operations.
- 3. Control Unit:** This unit is the back bone of computers. It is responsible for coordinating tasks between all components of a computer system.
- 4. Memory Unit:** Once a user enters data using input devices, the computer system stores this data in its memory unit.

**3.Primary Memory:** Primary memory is internal memory of the computer. It is also known as main memory.

**Types of Primary Memory– Primary memory is generally of two types.**

**1. RAM 2. ROM**

**1. RAM (Random Access Memory):**

1. It stands for Random Access Memory. RAM is known as read /write memory.
- 2.It is generally referred to as main memory of the computer system.
3. It is a temporary memory. The information stored in this memory is lost as the power supply to the computer is switched off.
- 4.That’s why RAM is also called as “Volatile Memory”.

**2. ROM (Read Only Memory):**

1. It stands for Read Only Memory. ROM is a Permanent Type memory.
- 2.The content is not lost when power supply is switched off.
- 3.Content of ROM is decided by the computer manufacturer and permanently stored at the time of manufacturing.
- 4.ROM cannot be overwritten by the computer. It is also called “Non-Volatile Memory”

**4. Secondary Memory:** It is an external memory of the computer. It is used to store the huge amount of different programs and information. The secondary storage devices are: **1.** Magnetic (Hard) Disk **2.** Magnetic Tapes **3.** Pen Drive **4.** Flash memory **5.** Optical Disk (CD, DVD) **6.** SSD etc.

**5. Output Unit:**

1. The third and final component of a computer system is the output unit.
2. After processing of data, it is converted into a format which human can understand.
3. After conversion, the output unit displays this data to users.
4. Examples of output devices include monitors, screens, printers and speakers etc. Thus, output units basically reproduce the data formatted by the computer for user's benefit

**4. Distinguish between RAM and ROM.**

ANS:

**RAM**

**ROM**

RAM stands for Random Access Memory.

ROM stands for Read Only Memory.

RAM is also called Temporary Memory.

ROM is called as permanent memory.

It is known as volatile memory.

It is known as non-volatile memory.

Types of RAM are: Dynamic RAM and Static RAM.

Types of ROM are: PROM, EPROM and EEPROM.

**1. Distinguish between GUI & CLI.**

ANS:

## GUI

GUI stands for graphical user interface.

Windows is characterised as GUI.

GUI uses pictures, symbols and word that control with mouse.

GUI is very friendly and easy to remember.

## CLI

CLI stands for command line interface.

DOS is characterised as CLI.

CLI uses characters on screen that control with keyboard.

CLI can be confusing and difficult to remember.

### 5. State the characteristics of Linux.

ANS: The characteristics of Linux are:

- I. Linux is Free and Open source.
- II. Linux Operating System is free from viruses.
- III. Easy to keep all the software up-to-date.
- IV. Linux is flexible and easily customisable.
- V. Linux provides excellent support.

### 6. State the different careers in IT.

ANS: The different careers in IT are:

- I. Web Designer and Developer- One can develop interactive websites using knowledge of HTML, PHP and various other programming languages.
- II. Software Developer- It is possible to have career as a software developer with the scripting skills.
- III. Database Manager- After getting knowledge about Database management one can work as a Database Manager with the help of SQL skills.
- IV. Professional Accountant- Using computerized accounting software one can successfully handle accounting work of any small or large scale organization.
- V. Financial Advisor- After getting IT knowledge one can advise and guide others in investing the money in various investment schemes with the calculations and benefits in future.

### 7. Name and explain the recent trends in IT.

ANS: The recent trends in IT are:

- I. Green Computing: It is the study and practice of environmentally sustainable computing or IT.
- II. The goals of green computing are similar to green chemistry:
  - a. To reduce the use of hazardous materials.
  - b. To maximize energy efficiency during the product's lifetime.
  - c. To recycle or biodegradation of non-functional products and factory waste. Green computing is important for all classes of systems, ranging from handheld systems to large-scale data centres.

### 8. State the functions of operating system.

ANS 1. A computer would not be able to function correctly without an operating system.

2. An operating system is a software program that empowers the computer hardware to communicate and operate with the computer software.
3. The operating system boots up the computer and makes sure that everything is operational.
4. The operating system is also what runs the cell phone and most of the electronic devices.

### 9. State the various operating systems.

ANS 1. Windows: Windows is a series of operating systems that are designed by Microsoft.

2. Mac OS: Apple Corporation's registered operating system is called Mac OS.
3. Chrome OS: Chrome OS is an open-source operating system created by Google.

4. UNIX: UNIX was trademarked in 1969 by a group of AT&T employees at Bell Labs

### **10. Explain different operating systems for mobile phone.**

- ANS 1. Android: Android is a free and open source operating system provided by Google. It is most popular OS amongst all other OS.
2. Asha: Asha is used by Nokia phones. This is closed source OS.
  3. Blackberry: This is a closed source operating system for smartphone and tablet devices.
  4. iOS: iOS is Apple's closed source operating system for Apple's iPhone, iPod Touch, iPad and second-generation Apple TVs.

### **11. Write a short note on GUI and CLI.**

- ANS1. GUI stands for graphical user interface.
2. Windows is characterised as GUI. This GUI is called GNOME
  3. GUI uses pictures, symbols and word that control with mouse.
  4. GUI is very friendly and easy to remember.

1. CLI stands for command line interface. The command line interface is when user interacts with the computer using text.
2. MS-DOS is characterised as CLI.
3. CLI uses characters on screen that control with keyboard.
4. CLI can be confusing and difficult to remember.

### **12. Write a short note on LAN MAN WAN.**

- ANS – 1. LAN covers smaller geographical area (Size is limited to a few kilometres) and are privately owned.
2. Usage area is limited to areas such as an office building, home, hospital, schools, etc.
  3. It covers a short distance, and so the error and noise are minimized.
  4. LAN is easy to setup.

**MAN** is larger area than that of a LAN and smaller area as compared to WAN.

1. It connects two or more separate computers that reside in the same or different cities.
2. It covers a large geographical area and may serve as an ISP (Internet Service Provider).
3. It is hard to design and maintain a Metropolitan Area Network.
4. Examples of a MAN are the part of the telephone company network that can provide a high-speed DSL line to the customer or the cable TV network in a city.

- WAN-**
1. Wide Area Network is a computer network that extends over a large geographical area.
  2. It might be confined within the bounds of a state or country.
  3. A WAN could be a connection of LAN connecting to other LAN's via telephone lines and radio waves.
  4. The technology is high speed and relatively expensive.

### **13. State the benefits of using LINUX.**

- ANS 1. Linux is Free and Open Source and Excellent Support operating system.
2. Linux Operating System is free from viruses.
  3. Easy to keep all the software up- to-date.
  4. Linux is flexible and easily customizable.

### **14. State different protocols used for internet.**

- ANS. 1. **TCP/IP:** Transmission Control Protocol & Internet Protocol. It breaks down the message into packets and sends them out into the network.
2. **DNS:** Domain Name System-It translates network address (such as IP addresses) into terms understood by humans (such as Domain Names) and vice-versa.

3.**DHCP:** Dynamic Host Configuration Protocol - It can automatically assign internet addresses to computers and users.

4. **FTP:** File Transfer Protocol - A protocol that is used to transfer and manipulate files on the internet.

**15.Explain different IT enabled services:**

ANS 1. Call Centres and Electronic Publishing

2.Medical Transcription and Data Centres

3.GIS Mapping and (Geographic Information System)

4. ERP (Enterprise Resource Planning) and Knowledge Management & archiving.

**16. Write a short note on:**

**1. Client-Server Architecture:** This type of architecture is most suitable for larger network. So there are two types of machines in network, client and server.

Client: A computer which is seeking any resource from another computer is a client computer. For

Example: Downloading an image file from a website, browsing Intranet/Internet etc. The network user normally uses a client computer to perform his day to day work.

Server: If a computer has a resource which is served to another computer, it is a server computer. The client establishes a connection to a server and accesses the services installed on the server.

for a network user to browse in internet or do spreadsheet work. For Example: Multimedia server, File storage servers, Webserver etc.

**2. Peer-to-Peer Architecture:**

1.In this type of architecture, all the machines, called as “peers”, have the same status and they can communicate with any other peer directly.

2. A Peer-to-Peer network has no dedicated servers. Its implementations are meant for small networks.

3. For Example, when students create a network to share files through Bluetooth or SHARE it, the mobile device forms peer-to-peer network for transfer of files.

**3.Internet:**

1.Internet means connecting computer to any other computer anywhere in the world. Internet is the highway of information.

2.The Internet has one very simple job: to move computerized information (known as data) from one place to another.

3.The Internet is generally defined as a global network connecting millions of computers.

4.The Internet is a massive network of networks, a networking infrastructure. With the help of the internet one can easily be in touch with anyone in the whole world by sending electronic mail, by chatting. Travel bookings can be made very easily; one can order books or buy anything online.

**4.History of Internet:**

1.The first workable prototype of the internet came in the late 1960's with the creation of ARPANET, or the Advanced Research Projects Agency Network. Originally funded by the U.S. Department of Defence,

2. ARPANET used packet switching to allow multiple computers to communicate on a single network. The technology continued to grow in the 1970's after scientists Robert Kahn and Vinton Cerf developed Transmission Control Protocol and Internet Protocol, or TCP/IP, a communications model that set standards for how data could be transmitted between multiple networks.

3.ARPANET adopted TCP/IP on January 1, 1983, and from there researchers began to assemble the “network of networks” that became the modern Internet.

4.The online world then took on a more recognizable form in 1990, when computer scientist Tim Berners-Lee invented the World Wide Web (WWW).